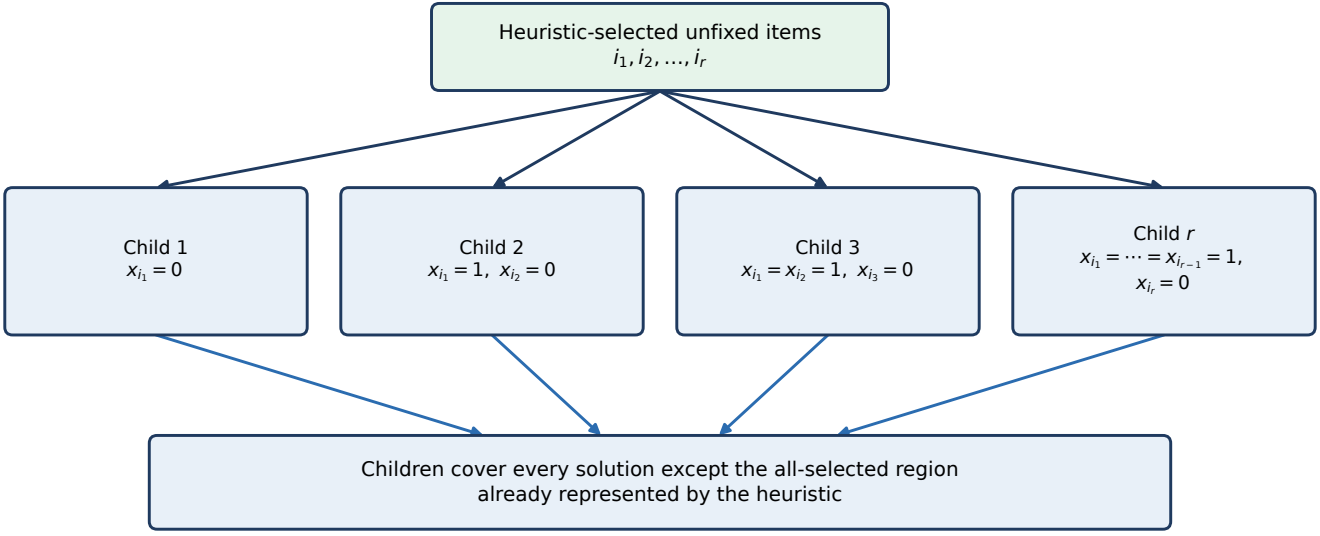
Heuristic-selected unfixed items
$i_1, i_2, \ldots, i_r$
Child 1
$x_{i_1} = 0$
Child 2
$x_{i_1} = 1,\ x_{i_2} = 0$
Child 3
$x_{i_1} = x_{i_2} = 1,\ x_{i_3} = 0$
Child r
$x_{i_1} = \cdots = x_{i_{r-1}} = 1,$
$x_{i_r} = 0$
Children cover every solution except the all-selected region
already represented by the heuristic